

VIRGINIA TECH

System Concepts Review

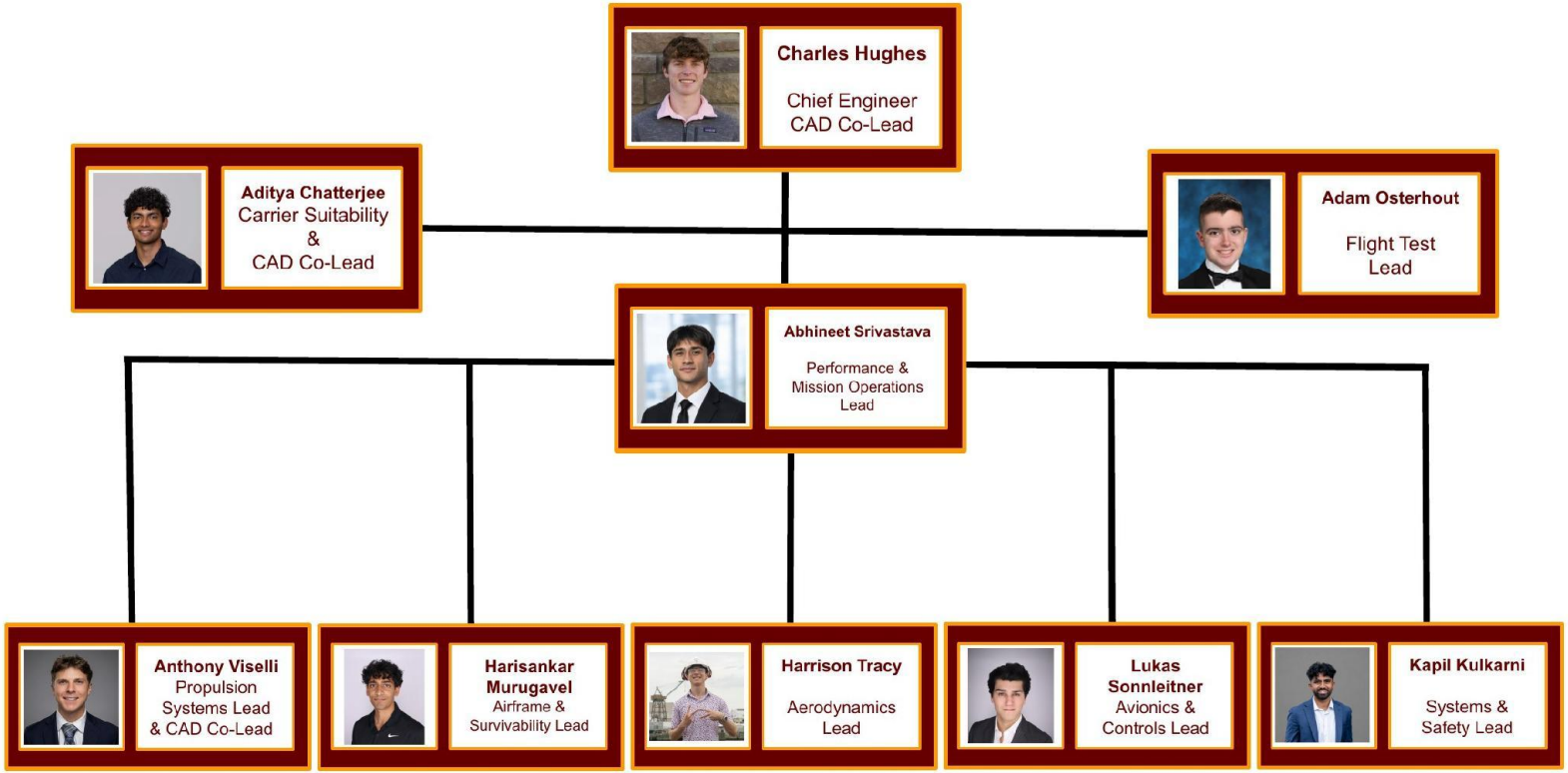
Gobbleworks | Team 11

NAVAIR Collaborative Combat Aircraft (CCA)

AOE 4065 · *October 16th, 2026* · *08:30 EDT* · *Virginia Tech*

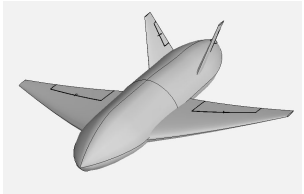
Presenter: Charles Hughes

Team Organization and Responsibilities

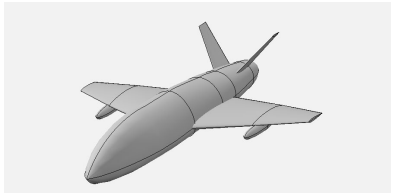


BLUF: Bottom Line Up Front

Bottom line: *Gobbleworks has designed 4 distinct CCA models, all of which respond to the carrier-based CCA RFP and focus on a strike mission, with plans to select one to move forward with by the CoDR.*



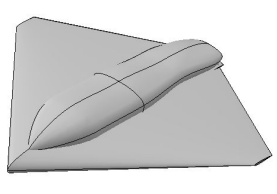
A: *X-1*
Reduced RCS



B: *X-2*
Ease of manufacturing and reduced drag



C: *X-3*
Volumetric Efficiency



D: *X-4*
Lowest RCS

WHAT WE WILL SHOW YOU

1 Requirements analysis
What problem are we solving?

2 Design strategy
What makes a good design?

3 Concepts
What different aircraft could solve it?

4 Project plan
How do we get from SCR to CoDR?

Origin of the Problem

THE OPERATIONAL PROBLEM

- A lower-cost uncrewed aircraft that operates alongside crewed fighters in a supplemental combat role (RFP).
- Carrier based, flying both air-to-air and air-to-ground missions.
- The aircraft closes the gap between the operational capabilities of a single sortie, to the cost of increasing them.

CUSTOMER AND OPERATORS

- Customer: U.S. Navy (Fleet)
- Operators and enablers:
  - Carrier air wing aircrew, flight deck and ordnance crew, AIMD maintenance technicians.
  - Navy Program Office and NAVAIR airworthiness authority.

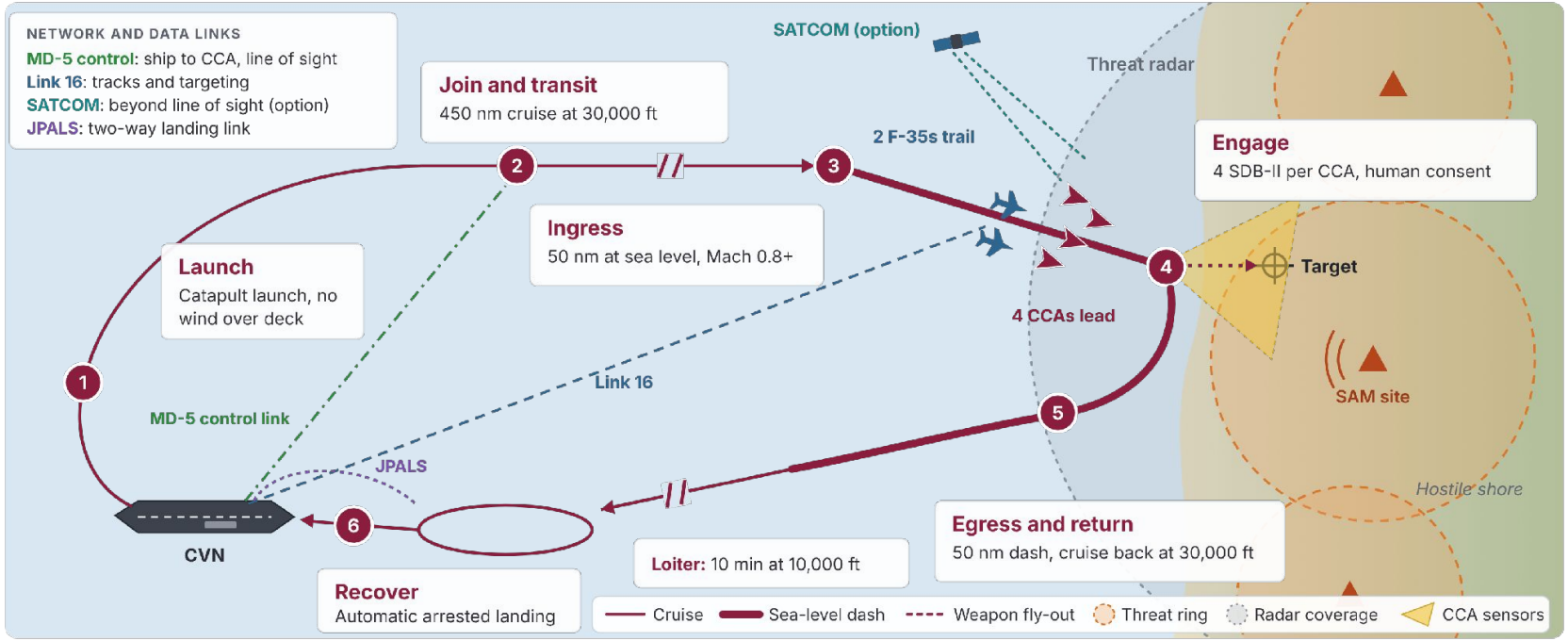
OPERATING ENVIRONMENT

- Catapult launch with wind over deck, arrested recovery, folded span 35 ft or less, spot factor 0.8 or less.
- Contested airspace; 10-year maritime service life.
- Communication between aircraft carrier, manned aircraft, and support crew (TBD).

WHY NOW: THE COST TREND

- The RFP sets out to break the Augustine's Law trend of rising combat-aircraft cost.
- At the $30M threshold the CCA costs about 40% of an F/A-18E/F and 27% of an F-35C.
- Air wing is able of carrying more munitions and supporting equipment per mission; cheaper than using another manned aircraft.

Concept of Operations



F2T2EA	Find	Fix	Track	Target	Engage	Assess
Kill chain across the package	CCA sensors search ahead	CCAs locate the target	Hold track, share on Link 16	F-35 crew assigns and approves	CCAs release 4 SDB-II each	Package checks effect on egress

Initial Operational Capability

2035

the customer's target IOC (RFP)

2035

IOC on our entry-into-service timeline
First unit delivery in 2034

Aircraft Units and Carrier's

The RFP requests a production run of 500 aircraft, compatible with CVN-68 and CVN-78 class aircraft carriers

Missions

Primary: Strike Mission

Ability to swap to Air-to-Air and EW packages

Carrier integration

Launch, automatic recovery and deck handling with existing ship systems (MD-5 station, JPALS).

Manufacturing and Sustainment

Target production rate: ~125 aircraft per year
Surge Capability: ~175 aircraft per year

Trained crews, shipboard maintenance and spares in place.

What IOC drives in the design: TRL 6 technologies and an existing production engine to keep development inside the schedule.

Comparator Aircraft

Aircraft	Carrier capable	MTOW	Radius / range	Internal bay	Unit cost	What it tells us
RFP / team target	Required	60,000 lb ceiling	500 nm radius (700 objective)	Desired	$30M threshold	Primary comparator aircraft is Northrop Grumman X-47B which tells us what needs to be modified to meet RFP requirements
Northrop Grumman X-47B	Yes	44,567 lb	>2,100 nm radius	Yes, 4,500 lb	~$400M	Only prior demonstration of autonomous carrier launch and recovery
Boeing MQ-25A Stingray	Yes	~44,000-lb class	500 nm radius	No (tanker)	~$200M	Only UAS operating in the carrier deck ecosystem today
Anduril FQ-44A Fury	No	Not published	~700 nm target radius	No	~$30M	Good cost per ability; not carrier suitable, below our size and no internal payload
Kratos XQ-58A Valkyrie	No	6,000–6,500 lb	>3,000 nm radius	Yes, 600 lb	~$4M	Best cost analog; not carrier suitable, far below our size and payload

Primary comparator: Northrop Grumman X-47B, because it is the closest technical analogue to our CCA requirements. The gap: no aircraft today pairs carrier-class capability with a CCA price point.

Where Our Requirements Come From

REQUIREMENTS BY SOURCE

38
RFP
System requirements, each traced to an RFP paragraph. Desired values (700 nm, Mach 0.95) are carried as objectives.

3
ConOps
Lost-link return or safe ditch (SUB-005); a person approves weapon release; deck handling with existing crew and gear

1
Market research
Unit cost of $30M or less at 500 aircraft, from USAF CCA Increment 1; Navy goal about $15M (SYS-039)

5
Standards
JSSG-2007D, MMPDS and ASTM B117, DoD TRL definitions, MIL-STD-1553C, NAVAIRINST 13034.1 with MIL-HDBK-516D

11
Engineering analysis
Sizing and structures flow-down: fuel fraction for radius, -1g - 5g and 200,000 lbf hook design case, catapult end speed, engine-out climb

3
Subsystem needs
JPALS receiver and 2-way datalink, one 1,000 lb mission-systems bay for both missions, weapons bay and station capacity

OWNED BY EVERY SUBTEAM

Discipline	Reqs
Chief Engineering & CAD	5
Carrier Suitability	10
Flight Test	4
Performance & Mission Ops	7
Propulsion	6
Airframe & Survivability	10
Aerodynamics	4
Avionics & Controls	9
Systems & Safety	6
Total (system + subsystem)	**61**

Evidence in handout: MathWorks .slreqx file (AVD012_Requirements), AVD-012 traceability matrix, AVD-009 stakeholder register, AVD-010 standards register.

Representative Requirements and Traceability

Source	System requirement	Derived / subsystem requirement	Owner	Verification
RFP 3.3.d Carrier suitability	**SYS-010** Two-way automatic carrier landing using differential GPS (JPALS)	**SUB-009** JPALS-compatible DGPS receiver and 2-way datalink integrated in the navigation subsystem	Avionics & Controls	Test, demonstration: simulated approaches, then at-sea demonstration
RFP 3.6.n Airworthiness	**SYS-038** Meet NAVAIR airworthiness requirements	**SUB-020** Design and verification follow the NAVAIRINST 13034.1 flight-clearance review and MIL-HDBK-516D criteria	Systems & Safety	Inspection, analysis: compliance matrix against MIL-HDBK-516D
RFP 3.5.2.a via THW3 sizing analysis	**SYS-023** Strike combat radius of 500 nm or more (700 nm desired)	**SUB-014** Usable fuel fraction sufficient for a 500 nm radius on both missions; top carpet-plot driver	Performance & Mission Ops	Analysis: mission sizing with calibrated drag polar and engine TSFC

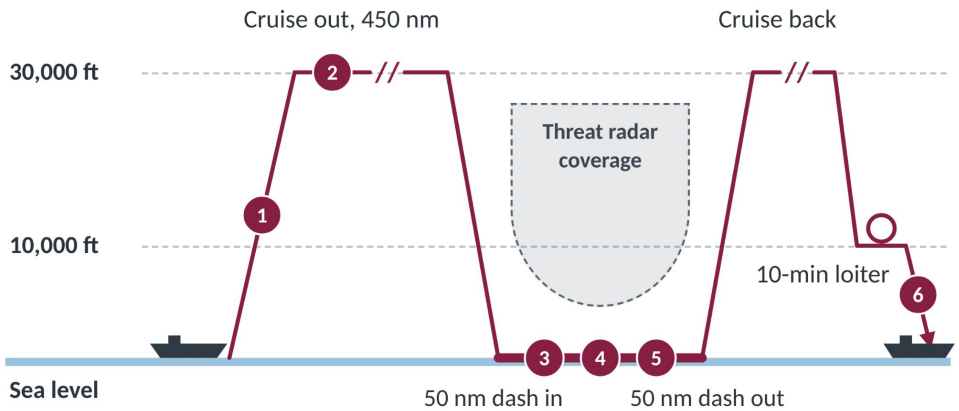
WHY THESE THREE

- SYS-010: with no pilot, recovery rests on avionics and the datalink; drives redundancy and reserved SWAP-C. Avionics, Carrier Suitability, Flight Test.
- SYS-038: sets the certification basis every subteam designs to, from structural margins to partitioned flight software.
- SYS-023: radius is the largest sizing driver, +27 to 47% TOGW from 500 to 700 nm. Performance, Propulsion, Aerodynamics, Airframe.

OPEN OR AMBIGUOUS

- CG sink at launch: the RFP invites trading more than 10 ft of sink for performance (SYS-012).
- Fuel for “two landing attempts” is not quantified (SYS-015).
- Does the 300 lb autonomy allowance count against the 1,000 lb avionics limit? (SYS-016, SYS-018)
- Resolved: spot factor is measured against the F/A-18E/F planform (SYS-008).

Design Mission and Performance Parameters



RFP: sea-level dash at Mach 0.8+, loiter at 10,000 ft. Team assumption: cruise at 30,000 ft. *Not to scale*

DESIGN MISSION

- Strike: 4× GBU-53/B SDB-II plus 1,000 lb of avionics, flown in 4-CCA packages.
- Alternate: air-to-air fighter sweep with 2× AIM-120D.
- Hard limits: launch at zero wind over deck, approach under 145 kt, 5 g at mid-mission weight, takeoff weight 60,000 lb or less.

Parameter	Threshold	Objective
Combat radius	500 nm	700 nm
Dash speed, air-to-air	Mach 0.8	Mach 0.95
Sea-level dash, strike	Mach 0.8	Mach 0.9
Combat at max thrust	2 min	5 min
Ordnance carriage	External acceptable	Internal
Weapons bring-back	50% of stores	Full load
Recovery reserve	10 min at 10,000 ft + 2 landing attempts	=Threshold
Spot factor	0.8 or less	Minimize
Unit cost, 500 aircraft	$30M	$22M

Design Drivers

CARRIER SUITABILITY

Low/High Speed

wind over deck at launch

Shapes wing area and high lift, wing fold, gear and hook structure (200,000 lbf hook load), deck footprint.

Measure of merit: Spot factor

RANGE

+47%

TOGW, 500 to 700 nm

Shapes fuel fraction, wing sizing and engine choice.

Measure of merit: Performance margin

ENGINE FUEL EFFICIENCY

+70 to 90 nm

Radius per 10% TSFC decrease

Installed vs Uninstalled TSFC. A 10% decrease also decreases TOGW by 10% to 15%.

Measure of merit: Performance margin, unit cost

MODULARITY

+22%

TOGW with strike plus EW

Shapes bay volume and modularity. Designing a larger bay to accommodate for future stores is also important.

Measure of merit: Modularity

STEALTH

+$5.75M

per aircraft vs external

Notably lower RCS. Shapes fuselage volume, bay doors and planform alignment.

Measure of merit: Stealth, ease of maintenance

AFFORDABILITY

$30M

threshold at 500 aircraft

Shapes materials, part count, engine choice and the level of low-observable treatment.

Measure of merit: Unit cost

Measures of Merit

Measure of merit	What it measures	Why it matters to the customer	How it will be evaluated	Weight
Unit Cost	Unit flyaway cost for 500 aircraft (Measured using combination of methods like DAPCA IV)	The RFP's binding cost deliverable	DAPCA IV with learning curve; threshold $30M	*.16*
Landing Stability Control	Stability margin and control-surface deflection required for carrier configuration	Landing on the carrier is essential to mission success	Stability derivatives during landing configuration	*.19*
RCS	How stealthy the aircraft is. How easily the aircraft is detectable to radar.	Flying with F-35s will require a threshold RCS to maintain stealth	1–5 proxy score, F/A-18E/F (low) to F-35C (high)	*.20*
Internal Weapons Carriage	How much of the weapons and ordnance can be carried internally	Carrying the weapons internally will allow for higher stealth and higher modularity	The amount of internal and external ordinance will be compared with the goal of 100% internal carriage	*.18*
Loiter Time	How long the aircraft can fly at low speeds	Longer loiter time may be critical while waiting for manned aircrafts and landing procedures	Trade studies using our sizing model	*.14*
Combat Radius	Mission range (nm)	Maximizing combat radius allows for a broader range of targets	Trade studies using our sizing models	*.13*

Requirement: a hard constraint that must be met. Measure of merit: a quantity we push past the requirement to separate compliant designs.

How the Customer Will Choose

1 — Combat value per dollar
"Maximum combat performance" within carrier suitability and cost limits.
Affordability

2 — Fits the air wing
"Minimal impact on operations for the carrier air wing"; no added risk to the carrier.
Modularity, Stealth, Maintenance

3 — Mission flexibility
"Cost-effective, mission flexible"; two payloads from one airframe, more considered.
Modularity, Unit Cost

4 — Low development risk
TRL 6 technologies and an existing production engine.
Unit Cost, maintenance

5 — Innovation
How does the proposal differ from existing products/aircraft.
All MoM's

WHO IS JUDGING

Navy program office

Cost-growth discipline and fleet integration.

External mentors (NAVAIR)

Based on PRMs 1 and 2, carrier ops, modularity, and stealth.

Course instructors

A traceable process: every decision sourced to the RFP or a named stakeholder, and design to quantifiable constraints.

What wins between two compliant designs

Determined by MoMs adherence.

How we will select the preferred system concept

1 Size	2 Analyze	3 Gate	4 Score	5 Decide
Size all four concepts to the same design mission with one sizing code.	Aerodynamics, propulsion, weights, performance, stability and control, cost.	Pass / fail against every requirement. Non-compliant concepts are fixed or dropped.	Weighted measures of merit, with a sensitivity check on the weights.	Team decision recorded in a decision memo; one PSC presented at CoDR.
ASAP	**By Nov 19**	**Nov 19–22**	**Nov 19–22**	**Nov 22, CoDR Dec 6**

FAIR COMPARISON RULES

- Same mission, margins and assumptions for every concept.
- Same tools and fidelity; methods owned by the functional teams.

SELECTION CRITERIA

- Gate: every requirement met.
- Rank: weighted MoMs, each with a measurable metric at CoDR (backup B5).

PROCESS AND RECORD

- Vote on winning concept based on data collected
- Document performance of various system concepts in respective ranking files.

Enabling Technologies

Technology	Design challenge it tackles	Potential benefit	TRL	Risk or limitation	Dependence
JPALS automatic carrier landing	Arrested recovery without a pilot (RFP 3.3.d)	Allows aircraft landing via GPS in all-weather and zero-visibility conditions	9	Needs a data link and can possibly jam in contested areas	High
Open mission systems (OMS/UCI, A-GRA)	Payload swaps and fleet integration	Common interfaces allow direct communication and payload swap across fleet	7	A-GRA isn't fully matured yet	High
Internal weapons bay and low observation shaping	Signature in contested airspace (RFP 3.1.e)	Lower radar signature and drag than external stores	9	Bay volume and weight compete with systems and fuel	Medium
Existing production engine, lower TSFC	Radius at lower takeoff weight (RFP 3.1.b.b)	Lower development cost and risk; lower fuel burn across mission	9	Thrust, size and TSFC fixed to what exists today	High
Composite airframe structure	Empty weight and maritime corrosion	Lower empty weight and corrosion resistance	9	Repair costs and damage inspection difficulty	Medium
Teaming autonomy software	Aircraft per operator; lost-link behavior	One operator controls several aircrafts; defined behavior if link is lost	6	Loss-link behavior must be proven	Medium

Key takeaway: JPALS is the technology we cannot do without; and at a TRL of 9 it is mature. RFP 3.1.d.a requires TRL 6; teaming autonomy software is closest to this floor.

Avionics & Software: Weight, Cost, and Code Size

AVIONICS WEIGHT

5.85%

of 17,105 lb empty weight baseline (Baseline from trade Study; 1,000 lb avionics bay, per RFP)

Basis: Within Raymer fighter range of 3–8% of empty weight (Table 11.6, via VT Module A7)

AVIONICS COST

$6.0M–$12.0M

per aircraft (2026 $) at 1,000 lb

Basis: $4,000–$8,000/lb (2012 $), Raymer Ch.18 / Nicolai & Carichner Ch.24 (VT Module A6); added separately from DAPCA IV cost study

SOFTWARE COST

TBD, as needed

COCOMO model can be used for rough estimation

Basis: (VT Module A6); GAO-22-105943: F-35 Block 4 cost grew $14.4B to $15.14B (2020–2021) — went over budget

LINES OF CODE

600K – 1.7M

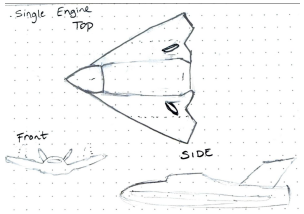
SLOC benchmark range: safety-of-flight to full avionics suite

Basis: 600,000 SLOC, MQ-25A safety-of-flight (Boeing, Jun 2026); 1.7M SLOC, F-22 avionics suite; cf. 9.5M on-board SLOC, F-35 (GAO-12-437)

Key takeaway: Our 1,000 lb avionics allowance sits inside the historical fighter range and should cost roughly $6–$12M per aircraft. Code size can vary depending on sophistication and extent of autonomy, but a good SLOC baseline is 600k from the MQ-25, and overall LOC between F-22 and F-35, as technology has become more sophisticated. Software cost is TBD pending an estimate at Conceptual Design Review (CoDR), if deemed necessary.

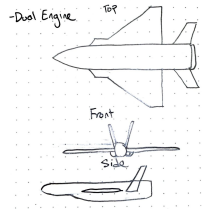
A Diverse Set of Candidate Concepts

8	1	7	4
candidates generated	screened out as infeasible	evaluated in the QFD	carried forward



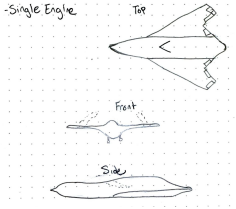
X1 - carried forward

+ *RCS reduction*
− Coupled yaw and roll



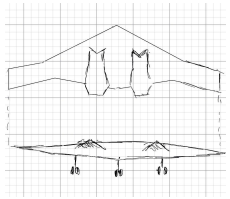
X-2 - carried forward

+ *Ease of manufacturing, reduced drag*
− Coupled yaw and roll



X-3 - carried forward

+ *Best volumetric efficiency*
− *Wing fold on a blended planform*



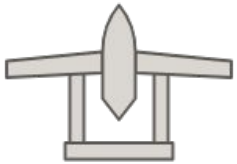
X-4 - carried forward

+ *Dual engine, lowest RCS*
− *Large spot factor*



X-5 Canard Delta - QFD

+ *High Trimmed lift*
− *Canard RCS and wake*



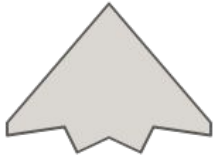
X-6 Twin-boom - infeasible

+ *Long loiter, low cost*
− *No mach 0.8 dash or 5g*



X-7 Lift-fan STOVL - QFD

+ *No catapult needed*
− *Lift system is heavy*



X-8 Tailless BWB (single engine) - QFD

+ *Best RCS*
− *No yaw authority*

From Requirements to Physical Solutions (RFLP)

R Requirement	F Function	L Logical alternatives	P Physical implementation
Provide thrust for takeoff, dash and combat within the 60,000 lb MOTW threshold, in production engine(s) only (RFP 3.3.1.b, 3.6.e-f)	Generate required thrust for each mission segment	*Single engine; Twin Engine*	Per concept: Engine model, inlet/nozzle design, engine placement
Carry strike and air-to-air ordnance; internal carriage desirable (RFP 3.1.e)	Carry, protect and release 4× GBU-53/B or 2× AIM-120D	External pylons; hybrid; internal bay	Per concept: bay location, door arrangement, rack type
Launch by catapult with zero-wind over the deck on a tropical day (RFP 3.3.1.b, 3.6)	Accelerate to flying speed within deck length and catapult stroke; transfer launch loads to the airframe	High thrust-to-weight ratio; High lift-wing; hybrid	Per concept: Wing planform area, nose bar, landing gear, max engine thrust (one engine)

QFD: Qualitative Evaluation of All Candidate Concepts

Measure of merit	X-47	X-1	X-2	X-3	X-4	X-5	X-6	X-7	X-8
Unit Cost	*0*	+	+	+	+	+	+	+	+
RCS	*0*	+	-	+	*0*	-	-	+	+
Internal Weapons Carriage	*0*	+	+	+	+	-	-	-	*0*
Landing Stability Control	*0*	-	+	+	+	*0*	-	-	*0*
Loiter Time	*0*	+	+	*0*	*0*	*0*	*0*	*0*	*0*
Requirement Concerns	*0*	*Stealth + aero performance*	*Stealth + engine inlet performance*	*External store capability + manufacturability*	*maneuverability + manufacturability*	*Canard RCS + Wake*	*Dash speed and maneuverability*	*Heavy lift system - Payload capacity*	*yaw authority without second engine*
Weighted Total	*0*	*+3*	*+3*	*+4*	*+3*	*-1*	*-2*	*0*	*+2*

Rate +, 0 or − against *the X-47* as the datum. Qualitative judgment.

MOST PROMISING, AND USEFUL TRADES

Internal Weapons Bay leads the design at the moment due to RCS, even though it is costlier (Modularity Trade Study, Stealth MoM)

UNSUITABLE, AND WHY

X5 - Small internal volume expected.
X6 - Shape is not RCS friendly. Low internal payload volume. Maintenance on tube structure.
X7 - Payload volume concerns, maintenance on lift system.
X8 - Yaw authority. No MoM, mission capability issue.

ASSUMPTIONS AND OPEN QUESTIONS

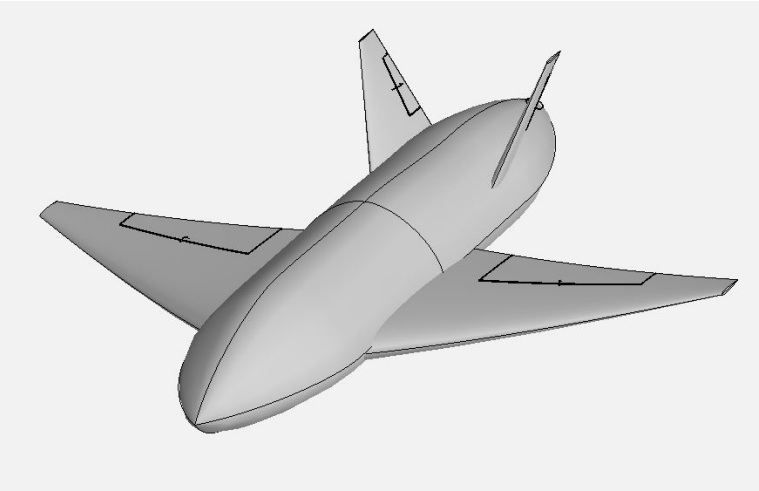
Ratings are assumed on high-level design impact of MoMs and RFP requirements rather than quantitative data at the moment.

OpenVSP models need refinement to quantify size constraints, and internal volume allowance.

Architecture and Configuration Choices

Decision	Concept A (Neet)	Concept B (Harrison)	Concept C (Hari)	Concept D (Adam)
Engine count	*Single*	*Single*	*Single*	*Dual*
Stores carriage	*Hybrid*	*Hybrid*	*Internal*	*Internal*
Planform / layout	*Conventional*	*Conventional*	*Delta/Blended Wing*	*Flying Wing*
Tail / control	*V-tail*	*V-tail*	*V-tail*	*No tail, differential drag*
Inlet / exhaust	*Bottom Fuselage*	*Bottom Center*	*Top Fuselage, low profile*	*Bottom/Side Fuselage*
Wing fold	*Yes*	*Yes*	*Yes*	*Yes*
Design Driver	*Stealth Based Design*	*Payload and Maneuverability*	*Aerodynamic Efficiency*	*Stealth*

Concept: X-1



WHY IT CAN DO THE MISSION

Single-engine F414 striker, 2000 lb hybrid carriage. Tapered mid-wing and V-tail on proven geometry, sized to close 5g and 700 nm radius without unconventional control or propulsion.

ADVANTAGES

- *V-tail: low RCS, low wetted area*
- *Hybrid carriage: clean for A2G, loaded for strike*
- *F414 is production hardware*

DISADVANTAGES

- *Ruddervators couple pitch and yaw*
- *Single engine, no redundancy overwater*
- *External stores cost radius*

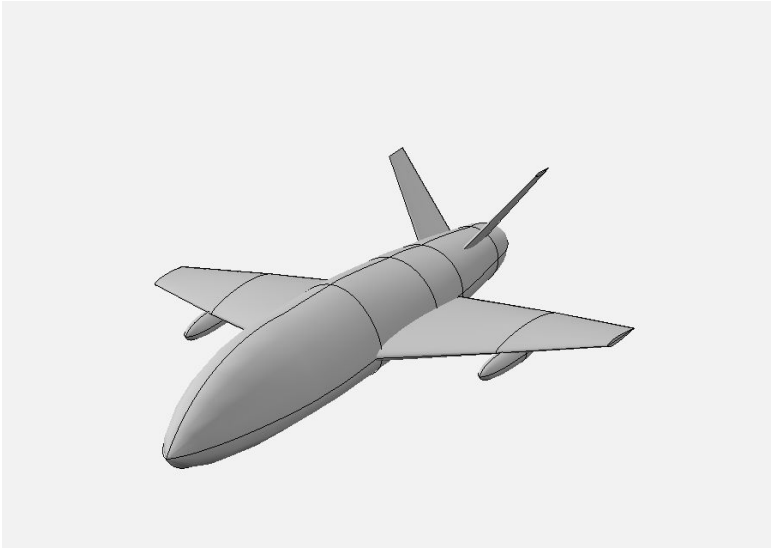
DISCRETE CHOICES THAT DEFINE THIS CONCEPT

Engines: *1*	Carriage: *hybrid*	Tail: *V-tail*
Planform: *Tapered mid-wing*	Inlet: *single*	Fold: *Yes*

RISKS AND UNKNOWNS

- *Tail sizing*
- *Spot factor ≤ 0.8 unverified*

Concept: X-2



DISCRETE CHOICES THAT DEFINE THIS CONCEPT

Engines: *1*	Carriage: *Hybrid*	Tail: *V-tail*
Planform: *Conventional*	Inlet: Center bottom	Fold: *Yes*

WHY IT CAN DO THE MISSION

Single-engine, conventional swept-wing CCA that would support the 500 nm radius and 0.8 dash with low technical risk, while folding wings and a compact V-tail keep it within carrier span and spot-factor limits.

ADVANTAGES

- *Aerodynamic design with small planform suitable for limited carrier space*
- *Hybrid weapon stores lower cost*

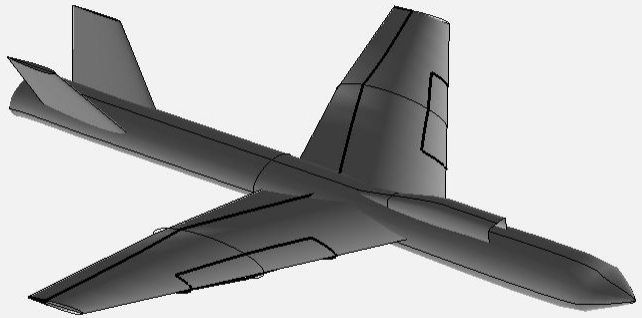
DISADVANTAGES

- *Bottom mounted inlet may impact RCS and internal weapon stores*
- *No engine redundancy, external weapons and fuel stores add to RCS*

RISKS AND UNKNOWNS

- *External weapon stores decrease stealth*
- *Smaller wings could result in higher takeoff velocity/lower lift*

Concept: X-3



DISCRETE CHOICES THAT DEFINE THIS CONCEPT

Engines: *1*	Carriage: *Internal*	Tail: *V-tail*
Planform: *Swept wing-body*	Inlet: *Top*	Fold: *Yes*

WHY IT CAN DO THE MISSION

A slender single-engine airframe with a swept wing and folding outer panels can meet range requirements, while a top inlet and no horizontal tail keep the shape clean for internal carriage.

ADVANTAGES

- *Stealth: no external stores; top inlet shielded from below.*
- *Spot factor: folding wing and slender body keep the stowed footprint small.*

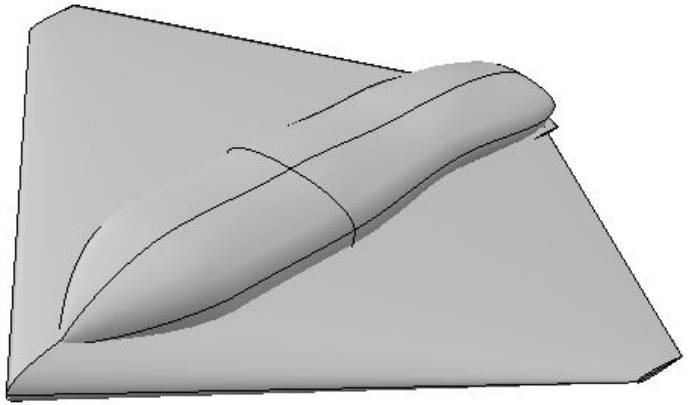
DISADVANTAGES

- *Modularity: slender body limits bay volume for kits.*
- *Carrier risk: one engine, no redundancy.*

RISKS AND UNKNOWNS

- *Does the strike load fit the bay in a slender fuselage?*
- *Inlet shielding; V-tail control at approach speed.*

Concept: X-4



WHY IT CAN DO THE MISSION

Combines high fuel and payload capacity with aerodynamic efficiency to support combat radius range and swept configuration to support dash speed of 0.8

ADVANTAGES

- *Low RCS, high payload, longer range*
- *Dual engine redundancy, increases survivability*

DISADVANTAGES

- *Less maneuverable, differential drag instead of rudder for yaw*
- *Difficult to manufacture, may underperform in a combat situation*

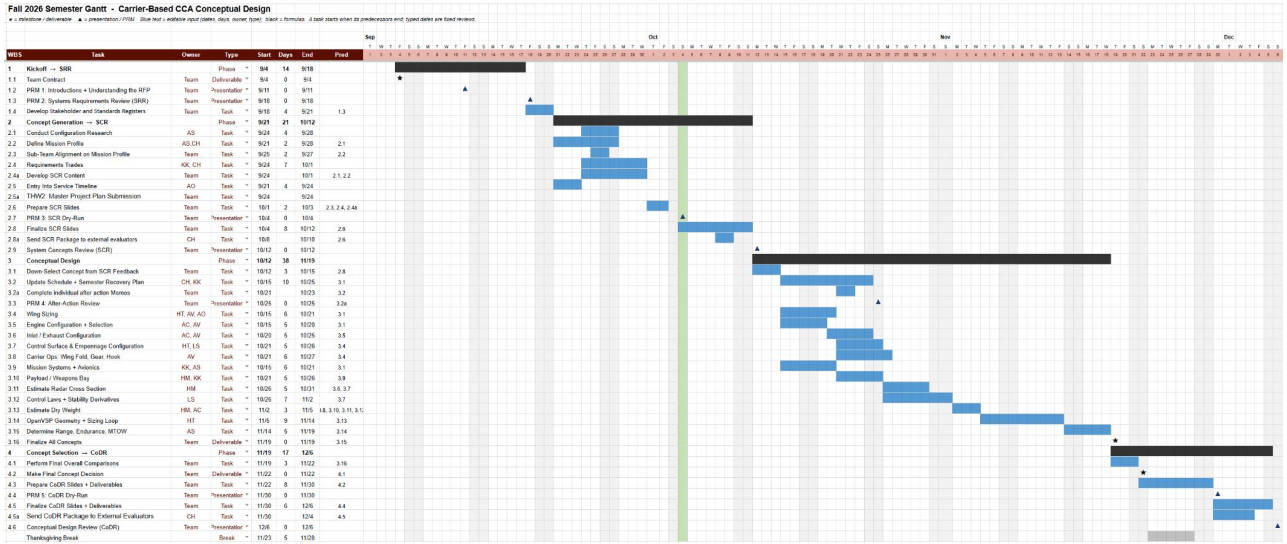
RISKS AND UNKNOWNS

- *Combat maneuverability is possibly at stake with this design*
- *Requires a complex control system for stability*

DISCRETE CHOICES THAT DEFINE THIS CONCEPT

Engines: 1/*2*	Carriage: *Internal*	Tail: *None*
Planform: *Flying wing*	Inlet: *Bottom Fuselage*	Fold: *Yes*

Schedule from SCR to CoDR



WBS	Task	Owner	Type	Start	Days	End	Pred
1	**Kickoff → SRR**		Phase	**9/4**	**14**	**9/18**	
1.1	Team Contract	Team	Deliverable	9/4	0	9/4	
1.2	PRM 1: Introductions + Understanding the RFP	Team	Presentation	9/11	0	9/11	
1.3	PRM 2: Systems Requirements Review (SRR)	Team	Presentation	9/18	0	9/18	
1.4	Develop Stakeholder and Standards Registers	Team	Task	9/15	4	9/21	1.3
2	**Concept Generation → SCR**		Phase	**9/21**	**21**	**10/12**	
2.1	Conduct Configuration Research	AS	Task	9/24	4	9/28	
2.2	Define Mission Profile	AS,CH	Task	9/21	2	9/28	2.1
2.3	Sub-Team Alignment on Mission Profile	Team	Task	9/25	2	9/27	2.2
2.4	Requirements Trades	KK, CH	Task	9/24	7	10/1	
2.4a	Develop SCR Content	Team	Task	9/24		10/1	2.1, 2.2
2.5	Entry Into Service Timeline	AO	Task	9/21	4	9/24	
2.5a	THW2: Master Project Plan Submission	Team	Task	9/24		9/24	
2.6	Prepare SCR Slides	Team	Task	10/1	2	10/3	2.3, 2.4, 2.4a
2.7	PRM 3: SCR Dry-Run	Team	Presentation	10/4	0	10/4	
2.8	Finalize SCR Slides	Team	Task	10/4	8	10/12	2.6
2.8a	Send SCR Package to external evaluators	CH	Task	10/8		10/10	2.6
2.9	System Concepts Review (SCR)	Team	Presentation	10/12	0	10/12	
3	**Conceptual Design**		Phase	**10/12**	**38**	**11/19**	
3.1	Down-Select Concept from SCR Feedback	Team	Task	10/12	3	10/15	2.8
3.2	Update Schedule + Semester Recovery Plan	CH, KK	Task	10/15	10	10/25	3.1
3.2a	Complete individual after action Memos	Team	Task	10/21		10/23	3.2
3.3	PRM 4: After Action Review	Team	Presentation	10/25	0	10/25	3.2a
3.4	Wing Sizing	HT, AV, AO	Task	10/15	6	10/21	3.1
3.5	Engine Configuration + Selection	AC, AV	Task	10/15	5	10/20	3.1
3.6	Inlet / Exhaust Configuration	AC, AV	Task	10/20	5	10/25	3.5
3.7	Control Surface & Empennage Configuration	HT, LS	Task	10/21	5	10/26	3.4
3.8	Carrier Ops: Wing Fold, Gear, Hook	AV	Task	10/21	6	10/27	3.4
3.9	Mission Systems + Avionics	KK, AS	Task	10/15	6	10/21	3.1
3.10	Payload / Weapons Bay	HM, KK	Task	10/21	5	10/26	3.9
3.11	Estimate Radar Cross Section	HM	Task	10/26	5	10/31	3.8, 3.7
3.12	Control Laws + Stability Derivatives	LS	Task	10/26	7	11/2	3.7
3.13	Estimate Dry Weight	HM, AC	Task	11/2	3	11/5	1.8, 3.10, 3.11, 3.12
3.14	OpenVSP Geometry + Sizing Loop	HT	Task	11/5	9	11/14	3.13
3.15	Determine Range, Endurance, MTOW	AS	Task	11/14	5	11/19	3.14
3.16	Finalize All Concepts	Team	Deliverable	11/19	0	11/19	3.15
4	**Concept Selection → CoDR**		Phase	**11/19**	**17**	**12/6**	
4.1	Perform Final Overall Comparisons	Team	Task	11/19	3	11/22	3.16
4.2	Make Final Concept Decision	Team	Deliverable	11/22	0	11/22	4.1
4.3	Prepare CoDR Slides + Deliverables	Team	Task	11/22	8	11/30	4.2
4.4	PRM 5: CoDR Dry-Run	Team	Presentation	11/30	0	11/30	
4.5	Finalize CoDR Slides + Deliverables	Team	Task	11/30	6	12/6	4.4
4.5a	Send CoDR Package to External Evaluators	CH	Task	11/30		12/4	4.5
4.6	Conceptual Design Review (CoDR)	Team	Presentation	12/6	0	12/6	
	Thanksgiving Break		Break	11/23	5	11/28	

CRITICAL PATH

3.1 → 3.4 → 3.7 → 3.12 → 3.13 → 3.14 → 3.15 → 3.16 → 4.1 → 4.2 → CoDR.

SCHEDULE RISK

CAD design validation within incorporated sizing model

Oct 16 — SCR

Oct 25 — PRM 4: after-action review

Nov 19 — All concepts sized

Nov 22 — Final concept decision

Nov 30 — PRM 5: CoDR dry run

Dec 6 — CoDR

N^2 Diagram: How the analyses depend on each other

Constraint and Mission Analysis	AR, Initial Wing Geometry, Foldable	Required Thrust	Aircraft Sizing Constraints		Mission Requirements		Payload Weight
	Aerodynamics	Sref, Cl, Cd, L/D	Wing Geometry, Cm, Cl, Cd	Cl, Cd, Wing Geometry	Wing Geometry	Wing Geometry, Cl, Cd, Cm	
		Propulsion	Fuel Consumed, MFF, TFF	Engine Location(s)		Fuel Consumed, TSFC	Flight Conditions, Fuel Weight
			Structures	Cg, Moments of Inertia	Landing Gear	Load Factor	Empty and Takeoff Weight
			Static Margin	**Stability and Control**		Tail Geometry, Trim Drag, Control Deflections	Static Margin and Maneuverability
			Avionics Volume and Payload Configuration		**Systems, Software, and Autonomy**	Weapons	
Range and Endurance						**Performance**	Range, Dash, Combat-Time
	Planform Area, Wetted Area	Flight Conditions	MTOW < 60,000 lb		Internal vs External Carriage		**Optimization**

Analysis module Feed-forward (read across a row) Feedback (below the diagonal)

N^2 coupling	What it drives
Aerodynamics gives L/D	Sizes thrust and structure
Propulsion gives fuel weight	Feeds structures and takeoff weight
Structures gives empty weight	Closes the loop to MTOW
Optimization gives MTOW	Capped below 60,000 lb

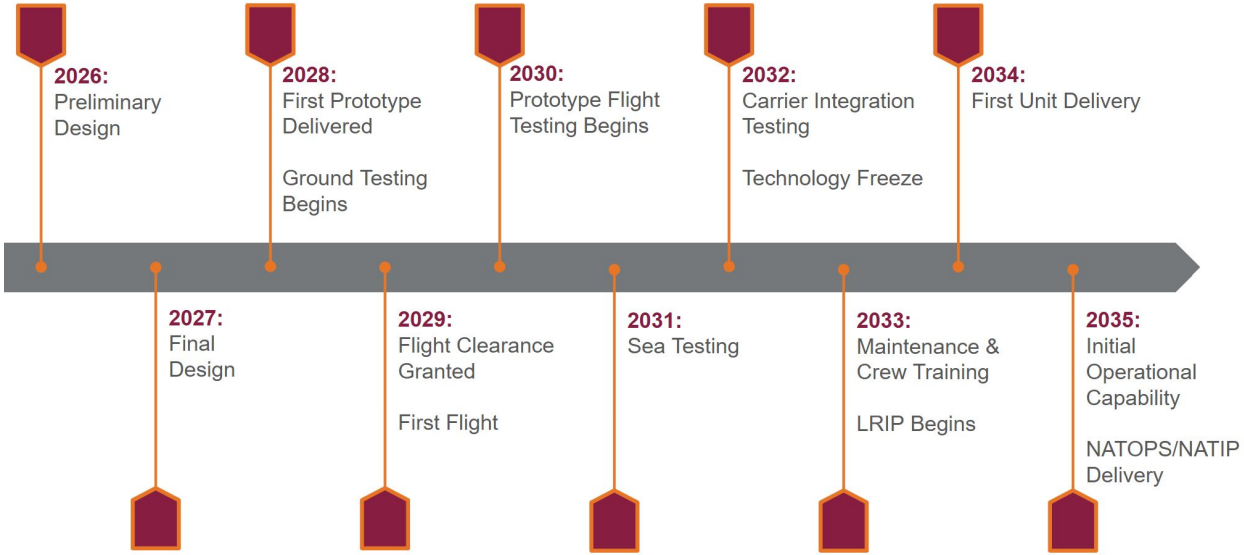
FEEDBACK AND ITERATION

Shaded cells below the diagonal loop back until takeoff weight and MTOW converge.

RFP DRIVERS IN THE LOOP

500 nm radius (700 desired), Mach 0.8 dash, MTOW under 60,000 lb, foldable wing.

Entry Into Service Timeline



Airworthiness obtained in compliance with NAVAIRINST 13034.1G . Interim flight clearance obtained prior to first flight.

Milestone	Date
ATP, detail design starts	2026
Prototype delivered	2028
Ground test	2028-2029
Flight test	2030-2033
LRIP start	2033
Certification complete	2034
First delivered, EIS	2034
IOC	2035

RFP IOC target: 2035.

Summary

1 REQUIREMENTS ANALYSIS

Origin of the problem and its operating environment [Slide 3–4]

Requirements from multiple sources [Slide 7–8]

The customer's IOC, explained [Slide 5]

A suitable comparator aircraft [Slide 6]

2 DESIGN STRATEGY

MoMs, mission profile, performance parameters [Slide 9–10]

Key design drivers, justified [Slide 11]

The customer's selection criteria [Slide 12]

Criteria and process for selecting the PSC [Slide 13]

3 CONCEPTS

A diverse set of viable concepts, pros and cons of each [Slide 17, 19–22]

Technologies for the major design challenges [Slide 14]

QFD across all candidate concepts [Slide 18]

4 PROJECT PLANNING AND ORGANIZATION

Organization chart, roles and responsibilities [Slide 23]

A timeline suited to project success [Slide 24]

Dependencies in the N^2 and the Gantt [Slide 25]

Entry-into-service timeline [Slide 26]



VIRGINIA TECH

Questions

Gobbleworks | Team 11

NAVAIR Collaborative Combat Aircraft (CCA)

Supporting material: backup slides and handouts (requirements file, traceability matrix, full HoQ, RFLP mappings, org chart, Gantt, N^2, EIS).

Presenter: All

Supplementals

B1 Requirements by source
B2 Stakeholder register
B3 Standards register
B4 Measure of merit detail (modularity)
B5 From SCR measures to CoDR criteria
B6 Supporting Trade Studies
B7 Responsibilities Matrix

B8 Acronyms
B9 X-4 Alternate Engine Configurations
B10 October Gantt Chart
B11 November Gantt Chart
B12 December Gantt Chart
B13 Testing Plan
B14 References
B15 References: Avionics & Software

B1 Requirements by source

ID	Requirement	Source	Owner	Verification
SUB-001	Engine is an existing production design	RFP 3.1.b.b; JSSG-2007A	Propulsion	*[Method]*
SUB-005	No additional risk to the carrier from autonomous operation	RFP 3.2; DoDD 3000.09	Systems & Safety	*[Method]*
SUB-008	Reserved SWAP-C to communicate with the MD-5 station and deck handling systems	RFP 3.3.c.b; MIL-STD-1553	Avionics & Controls	*[Method]*
SYS-038 / SUB-020	Meet NAVAIR airworthiness requirements	RFP 3.6.n; NAVAIRINST 13034.1; MIL-HDBK-516	Systems & Safety	*[Method]*

B2 Stakeholder register

Stakeholder	Interest	Design impact
Navy Program Office	Cost-growth discipline, fleet integration	Open Mission Systems (OMS/UCI), Link 16
NAVAIR Airworthiness / Cyber Authority	Certifiability	MIL-STD-1530 structures, partitioned flight-critical software
Flight Deck & Ordnance Crew	Deck cycle time, no unique GSE	Standard tow and tie-down fittings, deck-compatible fold time
Air Wing Aircrew	Trust and workload	Independent datalink, onboard collision avoidance
AIMD Maintenance Techs	Fast turnaround	Ergonomic LRU mass limit, stand-free access
Congress / OSD CAPE	Sustained affordability case	Cost-sensitivity reporting, minimized unique tooling
Engine OEM	No new qualification program	Constrained to published engine envelope
Course Instructor	Traceable process	Every requirement sourced to the RFP or a named stakeholder

B3 Standards register

Standard	Title / scope	Area	Where it applies
NAVAIRINST 13034.1	Flight clearance policy	Airworthiness	Flight clearance for the CCA
MIL-HDBK-516	Airworthiness certification criteria	Structures, flight technology, avionics	Airworthiness criteria (RFP 3.6.n)
DO-178C	Software considerations in airborne systems	Software	Flight software development
MIL-STD-1553	Digital time division command / response data bus	Avionics	Data bus; ground test certification
JSSG-2007A	Joint service specification guide: turbine engines	Propulsion	Existing production engine qualification
DoDD 3000.09	Autonomy in weapon systems	Autonomy	Human judgment over use of force
MIL-STD-882E	System safety	Safety	System safety certification
MIL-HDBK-244A	Aircraft stores compatibility	Payload	Stores certification
MIL-HDBK-2066	Catapulting and arresting gear forcing functions	Carrier suitability	Launch and recovery gear design
AEP-4671	UAV system airworthiness requirements	Airworthiness	UAV airworthiness

B4 Measure of merit detail: Modularity

DEFINITION

How many different missions one airframe can fly by swapping payload configurations, with no structural change or resizing. A mission counts only if the aircraft still makes the 500 nm combat radius.

WHY IT IS APPROPRIATE

The RFP asks for a "cost-effective, mission flexible" CCA with two payloads from one airframe and consideration of more. EW was emphasized by the NAVAIR mentors in the PRM meetings.

HOW IT IS MEASURED

Count of missions from air-to-air, strike, EW self-protection, decoy carriage and ISR. Threshold 2, target 3 to 4. Concepts rated 1–5 on weight margin, common interfaces, open software, power and cooling reserve, deck swap effort.

HOW IT RESPONDS TO CHANGING REQUIREMENTS

Pulls against internal carriage (bay volume caps every payload) and against tighter folded span and spot factor. A longer radius spends weight on fuel that kits would otherwise use.

B5 From SCR measures to CoDR selection criteria

Measure of merit at SCR	Selection criterion at CoDR	Unit	Method or tool	Weight
Unit recurring cost	Unit cost at 500 aircraft	$M	DAPCA IV with learning curve	*.14*
Spot factor	Folded planform area relative to fly-ready aircraft	ratio	Folded three-view from OpenVSP / CAD	*.15*
Stealth	Fraction of ordnance carried internally and geometry	RCS	Proxy score or open-source solver	*.2*
Modularity	Missions flown at 500 nm or more by payload swap	count	Sizing code rerun for each payload kit	*.14*
Ease of maintenance	On deck or below deck servicing	Time to service	Conceptual based on real carrier aircraft	*.1*
Performance margin	Combat radius; dash Mach; time at combat thrust	nm; Mach; min	Mission analysis with the common sizing code	*.1*

B6 Supporting trade studies

Study	Owner	Key result	Caveat
Wing geometry vs combat radius	H. Tracy	Baseline closest to 700 nm: AR 2.73, sweep 53.4°, t/c 0.105, cruise Mach 0.82	Historical regressions, constant TSFC
L/D vs combat radius	A. Osterhout, H. Tracy	500 to 700 nm needs cruise L/D of about 8 to 11	Not yet integrated with propulsion or weights
Operating empty weight model	A. Chatterjee	Closes at 29,028 lb takeoff weight, 17,105 lb OEW (0.59)	Component regressions carry 10–30% error
Stores carriage: cost and RCS	A. Chatterjee, K. Kulkarni	Internal adds $5.75M per aircraft, hybrid $3.01M; about 34 dB lower RCS than external	Placeholder RCS indices and labor multipliers
Mission Monte Carlo	A. Srivastava	Strike 0.991 vs air-to-air 0.965 weighted score; 4-CCA package meets 90% success	Notional Pk, threat rates and weapon costs
Strike payload and EW	L. Sonnleitner	Strike plus EW: takeoff weight +22%, unit cost +16% ($27.2M to $31.4M)	Mission segments not yet formalized
Carriage and EW (stealth)	H. Murugavel	Internal carriage +7.5% takeoff weight, +4.8% unit cost; EW is a major driver	Weight effects only; stealth scored, not computed
Takeoff weight vs radius and TSFC	A. Viselli	500 to 700 nm: +47% takeoff weight; 10% better TSFC saves 10–15%	Fixed L/D values, historical empty-weight fit
Engine survey	C. Hughes	14 engines; thrust required 10,698 lbf cruise, 7,845 lbf loiter; target TSFC 20% below F414	Preliminary aero and structures inputs

B7 Responsibilities Matrix

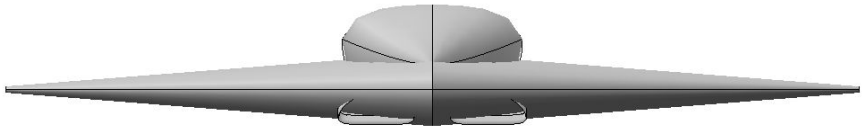
Functional team	Lead	Owns
Chief Engineering & CAD	Charles Hughes	Overall design, schedule, integration
Carrier Suitability & CAD	Aditya Chatterjee	Launch, recovery, deck fit
Flight Test	Adam Osterhout	Test and certification timelines
Performance & Mission Ops	Abhineet Srivastava	ConOps, mission modeling, performance
Propulsion Systems & CAD	Anthony Viselli	Engine selection, thrust and TSFC
Airframe & Survivability	Harisankar Murugavel	Airframe geometry, weights, RCS
Aerodynamics	Harrison Tracy	Planform, drag, inlet shaping
Avionics & Controls	Lukas Sonnleitner	Avionics, power, control surfaces
Systems & Safety	Kapil Kulkarni	Requirements, mission systems, safety

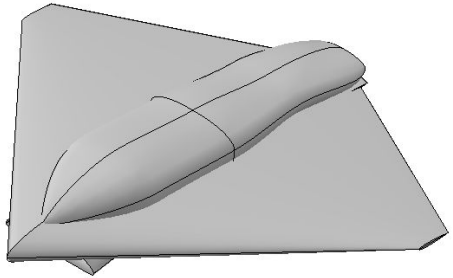
B8 Acronyms

Acronym	Definition
AIMD	Aircraft Intermediate Maintenance Department
ATP	Authorization to Proceed
CCA	Collaborative Combat Aircraft
CoDR	Conceptual Design Review
ConOps	Concept of Operations
DAPCA	Development and Procurement Costs of Aircraft
EIS	Entry Into Service
EW	Electronic Warfare
HoQ	House of Quality
IOC	Initial Operational Capability
JPALS	Joint Precision Approach and Landing System
LO	Low Observable
LRU	Line Replaceable Unit
MoM	Measure of Merit
MTOW	Maximum Takeoff Weight

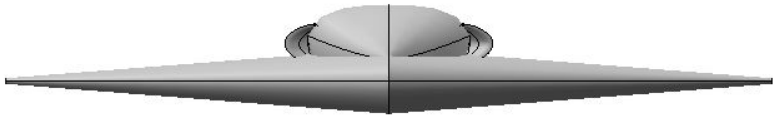
Acronym	Definition
OEW	Operating Empty Weight
OMS	Open Mission Systems
PSC	Preferred System Concept
QFD	Quality Function Deployment
RCS	Radar Cross Section
RFLP	Requirements, Functional, Logical, Physical
RFP	Request for Proposal
SCR	System Concepts Review
SDB	Small Diameter Bomb
SWAP-C	Size, Weight, Power and Cost
TOGW	Takeoff Gross Weight
TRL	Technology Readiness Level
TSFC	Thrust Specific Fuel Consumption
WBS	Work Breakdown Structure
WOD	Wind Over Deck

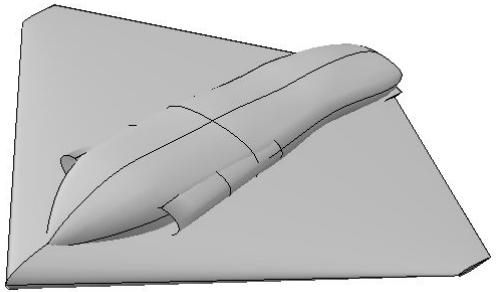
B9 X-4 Alternate Engine Configurations





Underside Intakes





Side Fuselage Intakes

B10 October Gantt Chart

October 2026

Month start Oct 1, 2026 **Month end** Oct 31, 202 **As of** Oct 4, 2026

Week columns: — phase ■ task ★ deliverable ▲ presentation/PRM ▒ break. Status is schedule-based.

WBS	Task	Owner	Type	Start	End	Days in month	Pred	Status	Wk Sep 28	Wk Oct 5	Wk Oct 12	Wk Oct 19	Wk Oct 26
2	**Concept Generation → SCR**		**Phase**	Sep 21	Oct 12	12		In progress	—	—	—		
2.4	Requirements Trades	KK, CH	Task	Sep 24	Oct 1	1		Complete	■				
2.4a	Develop SCR Content	Team	Task	Sep 24	Oct 1	1	2.1, 2.2	Complete	■				
2.6	Prepare SCR Slides	Team	Task	Oct 1	Oct 3	3	2.3, 2.4, 2.4a	Complete	■				
2.7	PRM 3: SCR Dry-Run	Team	Presentation	Oct 4	Oct 4	1		Due today	▲				
2.8	Finalize SCR Slides	Team	Task	Oct 4	Oct 12	9	2.6	In progress	■	■	■		
2.8a	Send SCR Package to external evaluators	CH	Task	Oct 8	Oct 10	3	2.6	Upcoming		■			
2.9	System Concepts Review (SCR)	Team	Presentation	Oct 12	Oct 12	1		Upcoming			▲		
3	**Conceptual Design**		**Phase**	Oct 12	Nov 19	20		Upcoming			—	—	—
3.1	Down-Select Concept from SCR Feedback	Team	Task	Oct 12	Oct 15	4	2.8	Upcoming			■		
3.2	Update Schedule + Semester Recovery Plan	CH, KK	Task	Oct 15	Oct 25	11	3.1	Upcoming			■	■	
3.2a	Complete individual after action Memos	Team	Task	Oct 21	Oct 23	3	3.2	Upcoming				■	
3.3	PRM 4: After-Action Review	Team	Presentation	Oct 25	Oct 25	1	3.2a	Upcoming				▲	
3.4	Wing Sizing	HT, AV, AO	Task	Oct 15	Oct 21	7	3.1	Upcoming			■	■	
3.5	Engine Configuration + Selection	AC, AV	Task	Oct 15	Oct 20	6	3.1	Upcoming			■	■	
3.6	Inlet / Exhaust Configuration	AC, AV	Task	Oct 20	Oct 25	6	3.5	Upcoming				■	
3.7	Control Surface & Empennage Configuration	HT, LS	Task	Oct 21	Oct 26	6	3.4	Upcoming				■	■
3.8	Carrier Ops: Wing Fold, Gear, Hook	AV	Task	Oct 21	Oct 27	7	3.4	Upcoming				■	■
3.9	Mission Systems + Avionics	KK, AS	Task	Oct 15	Oct 21	7	3.1	Upcoming			■	■	
3.10	Payload / Weapons Bay	HM, KK	Task	Oct 21	Oct 26	6	3.9	Upcoming				■	■
3.11	Estimate Radar Cross Section	HM	Task	Oct 26	Oct 31	6	3.6, 3.7	Upcoming					■
3.12	Control Laws + Stability Derivatives	LS	Task	Oct 26	Nov 2	6	3.7	Upcoming					■

B11 November Gantt Chart

November 2026

Month start Nov 1, 2026 **Month end** ov 30, 202 **As of** Oct 4, 2026

Week columns: — phase ■ task ★ deliverable ▲ presentation/PRM break. Status is schedule-based.

WBS	Task	Owner	Type	Start	End	Days in month	Pred	Status	Wk Oct 26	Wk Nov 2	Wk Nov 9	Wk Nov 16	Wk Nov 23	Wk Nov 30
3	**Conceptual Design**		**Phase**	Oct 12	Nov 19	19		Upcoming	—	—	—	—		
3.12	Control Laws + Stability Derivatives	LS	Task	Oct 26	Nov 2	2	3.7	Upcoming	■	■				
3.13	Estimate Dry Weight	HM, AC	Task	Nov 2	Nov 5	4	3.8, 3.10, 3.11, 3.12	Upcoming		■				
3.14	OpenVSP Geometry + Sizing Loop	HT	Task	Nov 5	Nov 14	10	3.13	Upcoming		■	■			
3.15	Determine Range, Endurance, MTOW	AS	Task	Nov 14	Nov 19	6	3.14	Upcoming			■	■		
3.16	Finalize All Concepts	Team	Deliverable	Nov 19	Nov 19	1	3.15	Upcoming				★		
4	**Concept Selection → CoDR**		**Phase**	Nov 19	Dec 6	12		Upcoming				—	—	—
4.1	Perform Final Overall Comparisons	Team	Task	Nov 19	Nov 22	4	3.16	Upcoming				■		
4.2	Make Final Concept Decision	Team	Deliverable	Nov 22	Nov 22	1	4.1	Upcoming				★		
4.3	Prepare CoDR Slides + Deliverables	Team	Task	Nov 22	Nov 30	9	4.2	Upcoming				■	■	■
4.4	PRM 5: CoDR Dry-Run	Team	Presentation	Nov 30	Nov 30	1		Upcoming						▲
4.5	Finalize CoDR Slides + Deliverables	Team	Task	Nov 30	Dec 6	1	4.4	Upcoming						■
4.5a	Send CoDR Package to External Evaluators	CH	Task	Nov 30	Dec 4	1	4.5	Upcoming						■
	Thanksgiving Break		Break	Nov 23	Nov 28	6		Upcoming						

B12 December Gantt Chart

December 2026

Month start Dec 1, 2026 **Month end** ec 31, 202 **As of** Oct 4, 2026

Week columns: — phase ■ task ★ deliverable ▲ presentation/PRM ▒ break. Status is schedule-based.

WBS	Task	Owner	Type	Start	End	Days in month	Pred	Status	Wk Nov 30	Wk Dec 7	Wk Dec 14
4	**Concept Selection → CoDR**		**Phase**	Nov 19	Dec 6	6		Upcoming	—		
4.5	Finalize CoDR Slides + Deliverables	Team	Task	Nov 30	Dec 6	6	4.4	Upcoming			
4.5a	Send CoDR Package to External Evaluators	CH	Task	Nov 30	Dec 4	4	4.5	Upcoming			
4.6	Conceptual Design Review (CoDR)	Team	Presentation	Dec 6	Dec 6	1		Upcoming	▲		

B13 Testing Plan

Expected 5-year test program with 4 test aircraft (2 ground test, 2 flight test)
Airworthiness compliant with NAVAIRINST 13034.1G, MIL-HDBK-516, and AEP-4671

Carrier Suitability
- Catapult Launch *(MIL-HDBK-2066)*
- Arrest Landing *(MIL-STD-882E)*
- Wing Folding

Structures
- Stores
- Loads
- Fatigue

Avionics
- Software *(DO-178C)*
- Power Systems
- Communication

Weapon Systems
- RCS
- Weapons Drop
- Live Fire
- Weapons Stores *(MIL-HDBK-244A)*

Performance
- Envelope Expansion
- Handling Qualities

Autonomy
- Autonomous Weapons Systems *(DoDD 3000.09)*
- Autonomous Control

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[6] C. Alberts, C. Wallen, D. Woody, and M. Bandor, "Managing Security and Resilience Risks Across the Lifecycle," Software Engineering Institute Blog, Carnegie Mellon University, Jul. 23, 2025. [Online]. Available: https://doi.org/10.58012/4apq-1606. [Accessed: Oct. 12, 2026].